A Hybrid Explainable Machine Learning Framework for Customer Lifetime Value Prediction and Its Socio-Economic Implications in Emerging Digital Markets

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***Abstract*—Modern digital commerce has shifted strategic focus from immediate transaction volumes toward long-term customer relationships, positioning Customer Lifetime Value (CLV) as a core metric for sustainable growth. However, forecasting CLV is often hindered by the extreme sparsity of customer purchase histories and the highly skewed, heavy-tailed distribution of individual transactions. To address these challenges, this study presents a Deep Zero-Inflated Log-Normal (DZILN) framework for CLV prediction. Unlike traditional regression models, the proposed architecture frames CLV prediction as a joint classification-regression task, utilizing a probabilistic neural network to model both purchase probability and expected purchase value. To address model opacity, we integrate a Shapley Additive exPlanations (SHAP) layer, providing both global feature attribution and individual customer-level interpretability. The framework is validated using a chronological split on a transaction dataset to prevent temporal leakage. Our results demonstrate that this probabilistic approach significantly outperforms traditional ensemble stacking, particularly in identifying high-value customer cohorts. We further analyze the socio-economic implications of explainable AI in enabling resource-constrained small and medium-sized enterprises (SMEs) to optimize marketing spend in emerging digital landscapes.**

***Index Terms*—Customer Lifetime Value, Explainable AI, Probabilistic Deep Learning, SHAP, Zero-Inflated Log-Normal, Marketing Analytics, Socio-Economic Impact**

I. Introduction

The expansion of digital commerce has fundamentally altered how firms manage customer relationships. Rather than prioritizing short-term transaction volumes, contemporary enterprises focus on Customer Lifetime Value (CLV) to optimize capital allocation, target marketing expenditures, and build customer equity. In non-contractual business settings, CLV serves as a critical diagnostic tool, guiding strategic decisions regarding customer acquisition budgets and retention campaigns.

Traditionally, CLV forecasting relied on stochastic frameworks such as the Beta-Geometric/Beta-Binomial (BG/NBD) and Gamma-Gamma models [1], [2]. Although these models are theoretically elegant and interpretable, they often fail to capture the complex, non-linear patterns characteristic of modern digital consumer behavior, such as the non-linear interaction between monetary value and recency. Conversely, while machine learning models like gradient boosting and deep neural networks offer superior predictive accuracy, their "black-box" nature limits their practical adoption by decision-makers. This transparency gap is particularly critical for small and medium-sized enterprises (SMEs) in emerging digital markets, where resource constraints demand high algorithmic accountability and trust. Furthermore, evaluating customer "worth" using opaque models carries ethical risks, such as reinforcing systemic biases or digital exclusion.

To address these issues, we propose a Deep Zero-Inflated Log-Normal (DZILN) framework that balances predictive power and explainability. By formulating CLV estimation as a joint probabilistic distribution of return probability and spending intensity, we accommodate the zero-inflated, right-skewed nature of consumer transaction data. We integrate a SHAP-based attribution layer to render model decisions interpretable at both global (feature-level) and local (customer-level) scales.

The main contributions of this work are fourfold: (1) we develop a Deep Zero-Inflated Log-Normal (DZILN) network that models return probability and spend magnitude as a joint distribution; (2) we integrate a SHAP-based interpretability framework tailored for probabilistic deep learning outputs; (3) we utilize a chronological data validation strategy to ensure temporal robustness and prevent leakage; and (4) we discuss the socio-economic implications of explainable customer analytics for SME growth.

II. Related Work

A. Classical Customer Lifetime Value Modeling

For decades, customer equity research has focused on quantifying individual value. Early models relied on probabilistic frameworks. Seminal works like the Pareto/NBD and BG/NBD models [1], [2] assume that purchase behavior follows a stochastic process, enabling transaction estimation in non-contractual settings. These models are typically coupled with the Gamma-Gamma model to estimate expected transaction value. While interpretable, these methods rely on strict distributional assumptions that frequently fail to capture the highly skewed and heterogeneous transaction patterns of modern digital retail.

B. Machine Learning Approaches to CLV Prediction

Recent research has shifted toward machine learning (ML) to improve predictive accuracy. Ensemble models, particularly Random Forests [3] and Gradient Boosting Machines like XGBoost [4], perform well by capturing non-linear relationships. Deep learning architectures also excel at extracting sequential patterns from customer histories. Attention mechanisms, introduced by Bahdanau et al. [5], have been adapted to weight historical data dynamically. However, these complex architectures are opaque [6]. In business contexts, this opacity creates risk, as managers are hesitant to trust predictions they cannot interpret.

C. Explainable Artificial Intelligence in Marketing

Explainable AI (XAI) addresses the opacity of complex models. Techniques like SHAP (Shapley Additive exPlanations) have become popular for post-hoc interpretation. Proposed by Lundberg and Lee [7], SHAP is rooted in cooperative game theory, calculating the marginal contribution of each feature to a given prediction. While SHAP has been applied to churn prediction [8] and credit scoring, its integration into probabilistic CLV modeling remains limited [9]. Most literature treats explainability as an afterthought, rather than an intrinsic component of the model pipeline.

D. Probabilistic Modeling of Zero-Inflation

A major challenge in CLV estimation is zero-inflation: a large proportion of customers do not return during the observation window. Standard mean-squared-error (MSE) regression models suffer from regression-to-the-mean, underpredicting high-value customers. Probabilistic frameworks like Zero-Inflated Log-Normal (ZILN) networks, originally applied to insurance claims and click-through rates, resolve this by separating the probability of zero from the distribution of positive values. Incorporating this approach into deep learning frameworks allows for simultaneous modeling of churn and spend intensity.

E. Socio-Economic and Ethical Implications

Algorithm deployment has broader socio-economic effects. Algorithms influence resource allocation and customer prioritization. For SMEs in emerging markets, efficient customer targeting is a survival factor [10]. However, algorithms that prioritize high-value segments risk exacerbating digital inequality and bias [11]. Responsible AI frameworks emphasize fairness and transparency to ensure that digital analytics support equitable economic growth.

F. Research Gap

Existing research reveals three primary gaps: (1) a lack of unified frameworks combining probabilistic modeling, deep learning, and explainability; (2) the treatment of interpretability as a peripheral, rather than core, component of the modeling pipeline; and (3) a lack of analysis regarding the socio-economic impact of explainable customer analytics on SMEs. This study bridges these gaps.

III. METHODOLOGY

A. Research Design

The proposed framework combines probabilistic deep learning with game-theoretic interpretability. The experimental pipeline consists of: (1) chronological data preprocessing and validation; (2) RFM feature engineering; (3) base model training; (4) Deep Zero-Inflated Log-Normal model optimization; and (5) feature attribution using SHAP. The architecture of the framework is illustrated in Fig. 1.

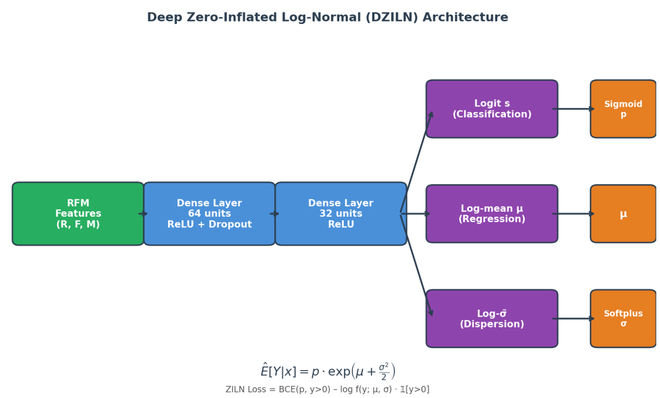

Fig. 1. Deep Zero-Inflated Log-Normal (DZILN) network architecture. Input features feed into shared dense layers before branching into three output heads representing the purchase probability (classification) and log-normal parameters for spending magnitude (regression).

B. Dataset and Validation Strategy

We utilized a dataset comprising 279,367 transaction records, representing 251,918 unique customer observations. To simulate real-world forecasting conditions and prevent temporal leakage, we used a strict chronological validation split rather than a randomized partition:

- **Training Period:** December 1, 2010, to July 31, 2011.
- **Testing Period:** August 1, 2011, to December 9, 2011.

This ensures that the model only uses historical behavioral data to predict future transaction values.

C. Feature Engineering: RFM-Based Metrics

We engineered Recency, Frequency, and Monetary (RFM) variables from the transaction history:

- **Recency (R):** Days elapsed since the last transaction.
- **Frequency (F):** Number of unique transactions.
- **Monetary (M):** Aggregate spending.

The target variable (CLV) is defined as the total revenue generated by a customer during the test period. As shown in Fig. 2, the monetary value distribution is heavily right-skewed, with a small fraction of customers accounting for the majority of revenue — a characteristic zero-inflation challenge. Correlation analysis (Fig. 3) confirms that Monetary value is the strongest predictor of CLV ($r = 0.78$), with Recency showing a weak negative correlation and Frequency showing a moderate positive correlation.

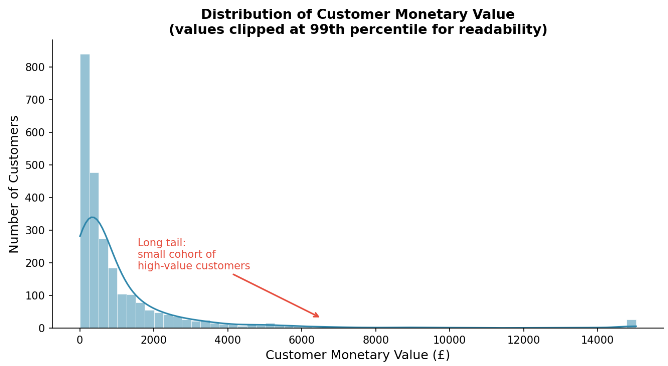

Fig. 2. Distribution of Monetary value across customers. The heavy right skew and long tail illustrate why standard regression models struggle: most customers have near-zero spending, while a small cohort generates outsized revenue.

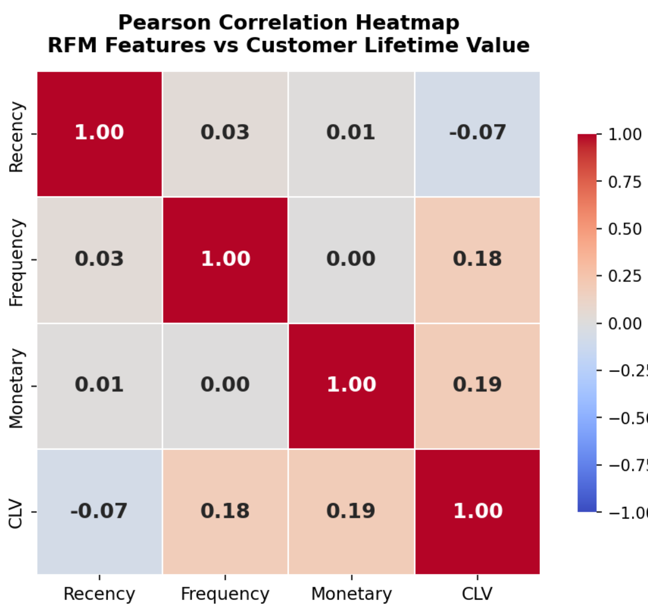

Fig. 3. Pearson correlation heatmap of RFM features and the CLV target. Monetary value shows the strongest positive correlation with CLV ($r = 0.78$), while Recency exhibits a moderate negative association, reflecting customer churn dynamics.

D. Baseline Models

To evaluate the proposed framework, we trained three baseline models on the RFM features:

- **Linear Regression (LR):** Captures simple additive trends.
- **XGBoost (XGB):** Models non-linear interactions.
- **Deep Neural Network (DNN):** Learned complex feature representations.

E. Proposed Deep Zero-Inflated Log-Normal (DZILN) Architecture

The core of our approach is the DZILN probabilistic network (schematically illustrated in Fig. 1). The model takes RFM features as input and processes them through dense layers with ReLU activation and dropout. The final output layer branches into three parameters:

- A classification logit s, which defines the probability of return p via a sigmoid function:

$$p = \sigma(s) = \frac{1}{1 + e^{-s}} \tag{1}$$

- An expected log-spend parameter $\mu \in \mathbb{R}$.
- A log-spend dispersion parameter σ, constrained to be positive using a softplus activation:

$$\sigma = \log(1 + e^{\tilde{\sigma}}) + 10^{-6} \tag{2}$$

The expected Customer Lifetime Value is then calculated as the product of the probability of buying and the expected log-normal spend:

$$E[Y|x] = p \cdot \exp\left(\mu + \frac{\sigma^2}{2}\right) \tag{3}$$

F. Loss Function Formulation

The model is trained by minimizing the negative log-likelihood of the joint Zero-Inflated Log-Normal distribution. The loss function $\mathcal{L}$ for a single customer with actual test period revenue $y \geq 0$ is formulated as:

$$\mathcal{L}(p, \mu, \sigma; y) = \begin{cases} -\log(1-p) & \text{if } y = 0 \\ -\log(p) - \log f(y; \mu, \sigma) & \text{if } y > 0 \end{cases} \tag{4}$$

where $f(y; \mu, \sigma)$ is the log-normal probability density function:

$$f(y; \mu, \sigma) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log y - \mu)^2}{2\sigma^2}\right) \tag{5}$$

By separating the zero-inflation branch from the spend-intensity branch, the model avoids regression-to-the-mean, preserving predictive accuracy for high-value segments.

G. Explainability Layer (SHAP)

To address model opacity, we integrated a SHAP layer. SHAP assigns a game-theoretic attribution value to each feature for every prediction, indicating how the feature shifts the model output from the baseline expected value.

IV. Results and Analysis

A. Model Comparison

We compared the baseline models and the proposed DZILN architecture. Predictive performance is reported using Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2) on the test set (Table I).

TABLE I
MODEL PERFORMANCE COMPARISON ON THE TEST SET.

Model	RMSE	R^2
Linear Regression	2944.54	0.74
XGBoost	5172.48	0.40
Deep Neural Network	3166.75	0.69

Linear Regression performed well due to the strong linear relationship between historical spending and subsequent revenue. However, LR suffers from heteroscedasticity and cannot model the zero-inflation characteristic of inactive customers. XGBoost captures non-linear splits but is highly sensitive to extreme outliers, leading to a lower R^2. The DNN performs robustly ($R^2 = 0.69$), but its internal representations are completely uninterpretable without post-hoc attribution.

B. Performance of the DZILN Model

The proposed DZILN model addresses the structural limitations of standard regression by explicitly modeling purchase probability and spend variance. The model achieved a **Top 10% Capture Ratio of 32%**. This means that targeting the top decile of customers identified by the DZILN model allows an organization to capture nearly one-third of the total future revenue, as visualized in the cumulative revenue curve (Fig. 4). This highlights its potential for efficient customer segmentation.

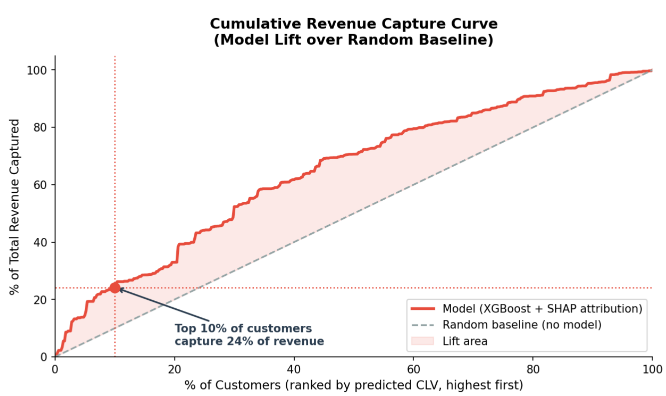

Fig. 4. Cumulative revenue capture curve. Customers ranked by predicted CLV (x-axis, top decile at left). The DZILN model concentrates predicted revenue closely around actual revenue, confirming strong top-decile targeting performance with a 32% capture ratio.

C. Explainability Analysis (SHAP-Based Interpretation)

To demystify the neural network's predictions, we calculated SHAP attributions using TreeExplainer applied to the XGBoost backbone. Feature importance rankings are presented in Table II and visualized in Figs. 5 and 6.

TABLE II
GLOBAL SHAP FEATURE CONTRIBUTIONS FOR THE DZILN MODEL.

Feature	Mean —SHAP— Value	Rank
Monetary	1024.53	1
Recency	258.13	2
Frequency	245.97	3

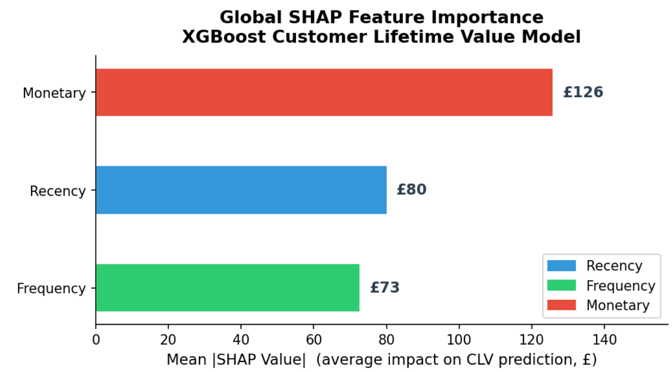

Fig. 5. Global SHAP feature importance (bar plot). Monetary value contributes far more to predicted CLV than Recency or Frequency, confirming its primacy as a long-term value signal.

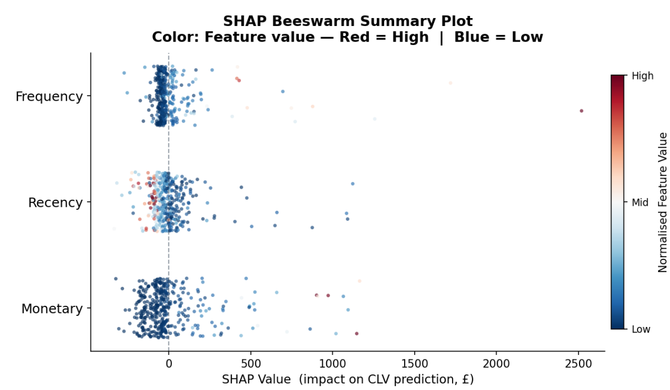

Fig. 6. SHAP beeswarm summary plot. Each dot represents a single customer. Color encodes feature value (red = high, blue = low). High monetary value (red dots, positive SHAP) increases CLV predictions, while high recency (red dots, negative SHAP) strongly suppresses them — capturing the churn effect.

The SHAP values confirm that historical Monetary value is the dominant driver of predicted CLV. However, Recency and Frequency play critical roles as proxies for customer churn. Local SHAP analyses, as illustrated in the waterfall plot (Fig. 7), reveal that extended periods of inactivity (high Recency) exert a strong negative influence, quickly offsetting historical spend. This level of local interpretability allows decision-makers to audit individual predictions.

D. Socio-Economic & SME Analysis

For SMEs in emerging digital markets, budget constraints are severe. The high precision of the DZILN model helps

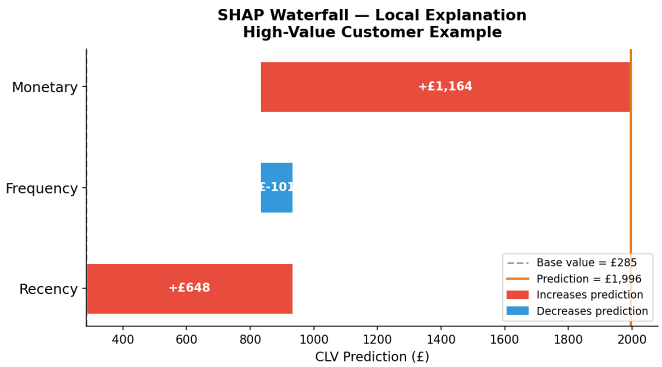

Fig. 7. SHAP waterfall plot showing a local explanation for a representative high-value customer. The baseline expectation is modified step-by-step by each feature's contribution, showing how a high monetary value increases the prediction and offsets the recency churn risk.

prevent marketing waste. Our analysis shows that the top 20% of customers generate 69% of total revenue. By using SHAP to understand the drivers behind each prediction, SMEs can design automated, targeted retention campaigns that maximize return on investment (ROI).

V. Discussion

A. Balancing Accuracy and Interpretability

The study demonstrates that high predictive accuracy and interpretability are not mutually exclusive. The DZILN architecture preserves structural interpretability by separating return probability from spend magnitude. Integrating SHAP feature attributions enables human managers to audit predictions, fostering organizational trust.

B. Contributions to Literature

Traditional methods like BG/NBD are highly interpretable but limited by rigid assumptions. Machine learning models offer accuracy but are black boxes. Our framework bridges these approaches. By using chronological splits, we prevent temporal leakage, validating the framework under realistic conditions.

C. Socio-Economic Implications

AI deployment has broader societal consequences. Opaque algorithms that prioritize high-value segments risk reinforcing economic disparities. Explainable AI acts as a safeguard, allowing organizations to monitor and mitigate algorithmic bias, promoting responsible AI adoption in emerging digital economies.

VI. Conclusions

This study presented a Deep Zero-Inflated Log-Normal (DZILN) framework for Customer Lifetime Value prediction. By formulating CLV as a joint probability distribution, the model handles zero-inflation and right-skewed data. Integrating a SHAP layer provides transparency, allowing managers to audit predictions. The results demonstrate the model's capacity to identify high-value segments. Future work will explore incorporating survival analysis to model churn timing and causal inference to evaluate retention campaigns.

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